

Quality of Object Localization and Detection

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Instructor:

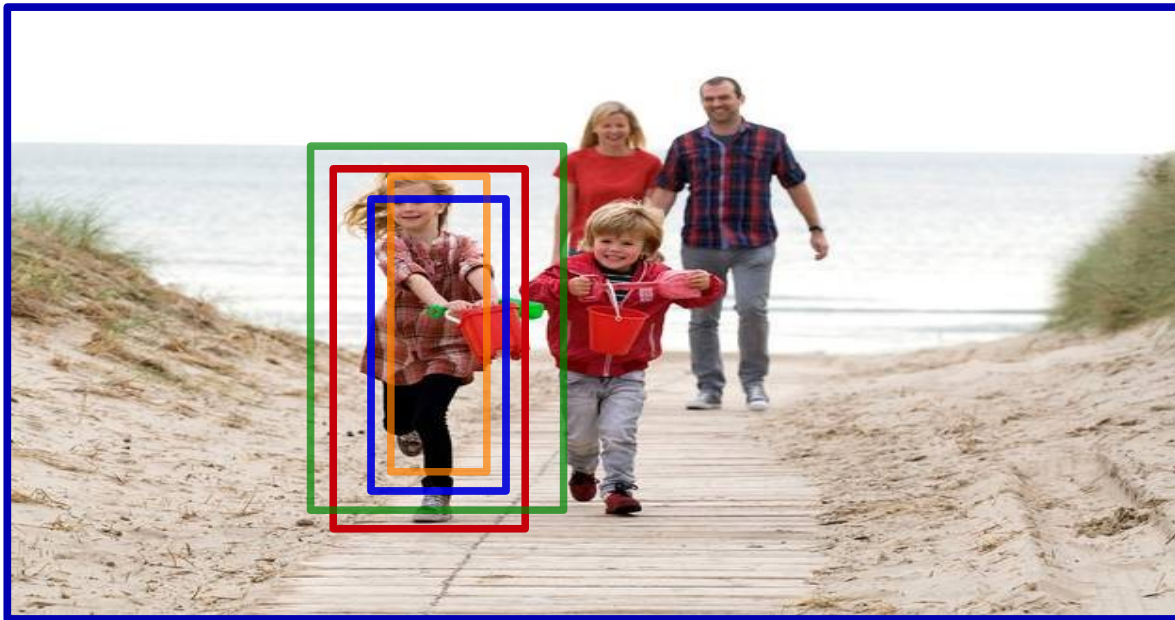
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Quality of Object Localization

Which Model is the Best ?



Ground Truth



Model #1



Model #2



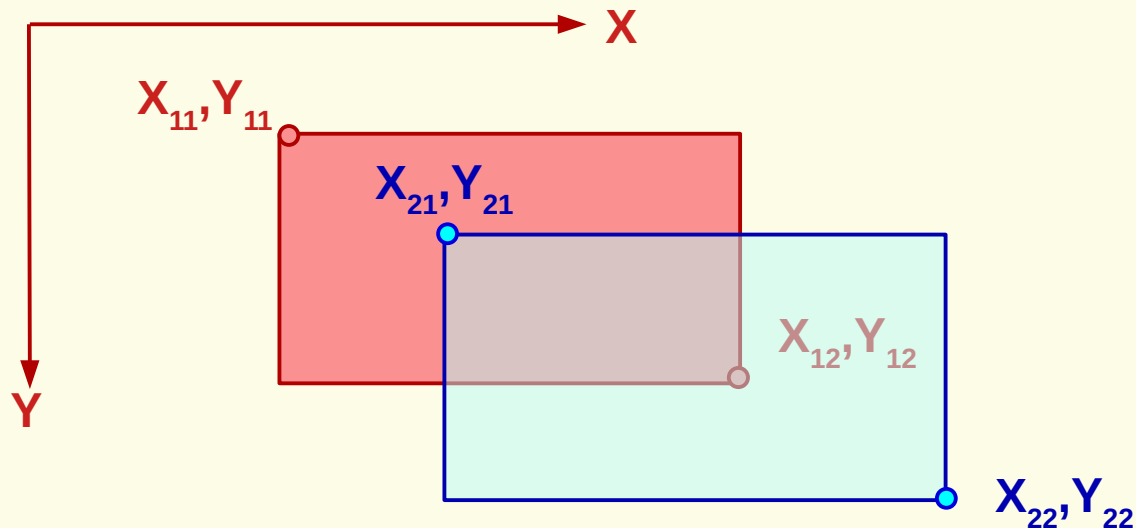
Model #3



Remember:

Each bounding box is defined by **X,Y,W,H** or **X₁ , Y₁, X₂, Y₂**

Regression Loss (Absolute or Squared Distance)



$$\text{Absolute distance} = |X_{11} - X_{21}| + |Y_{11} - Y_{21}| + |X_{12} - X_{22}| + |Y_{12} - Y_{22}|$$

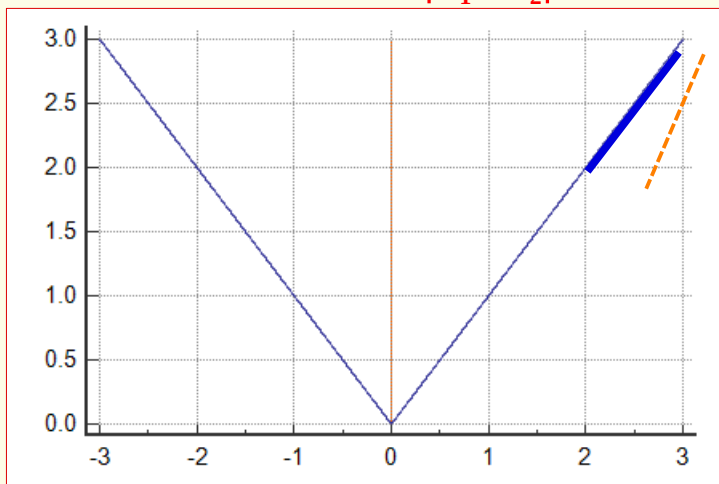
$$\text{Squared distance} = (X_{11} - X_{21})^2 + (Y_{11} - Y_{21})^2 + (X_{12} - X_{22})^2 + (Y_{12} - Y_{22})^2$$

Which is better: MSE or MAE

$$\text{Absolute distance} = |X_{11} - X_{21}| + |Y_{11} - Y_{21}| + |X_{12} - X_{22}| + |Y_{12} - Y_{22}|$$

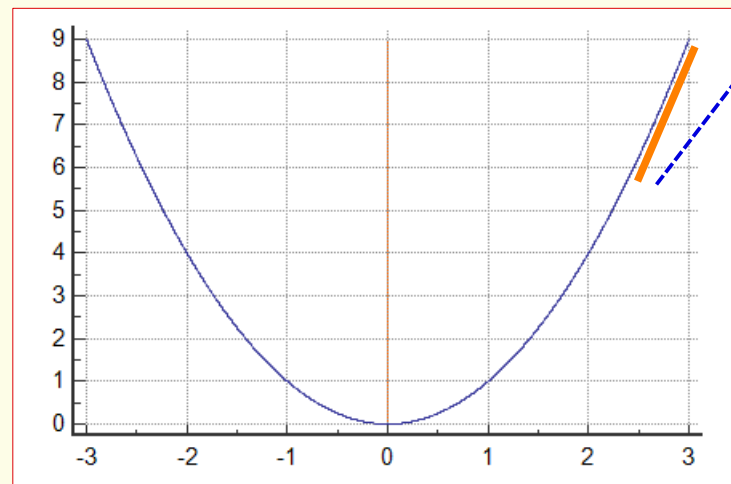
$$\text{Squared distance} = (X_{11} - X_{21})^2 + (Y_{11} - Y_{21})^2 + (X_{12} - X_{22})^2 + (Y_{12} - Y_{22})^2$$

Absolute: $|x_1 - x_2|$



Difference $(x_1 - x_2)$

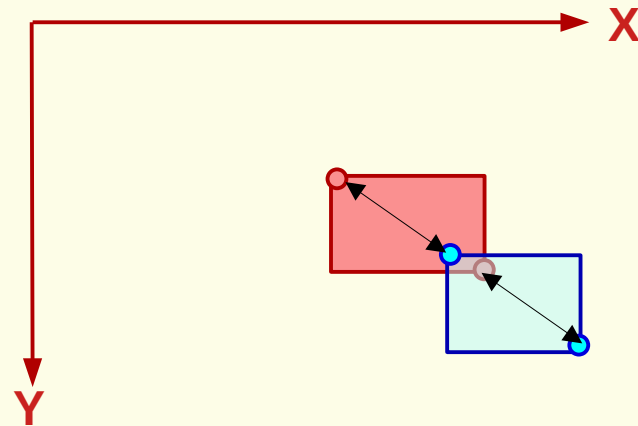
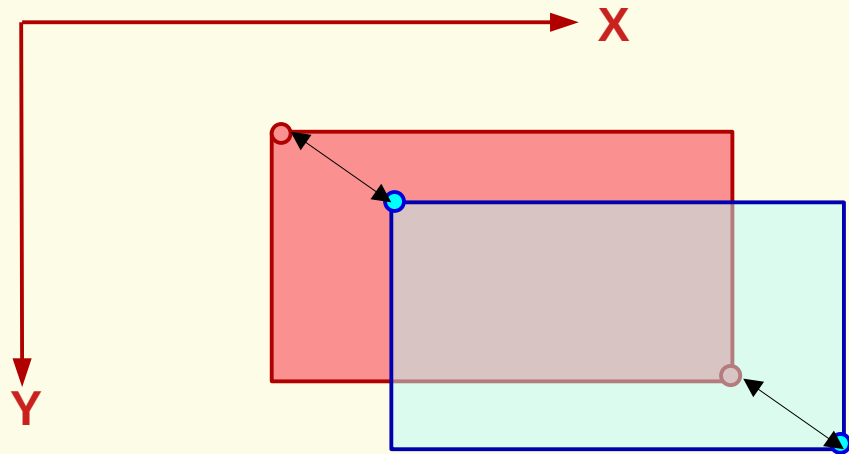
Squared: $(x_1 - x_2)^2$



Difference $(x_1 - x_2)$

As the **difference** increases,
Gradient of squared error increases rapidly than of Absolute error,
 **Faster Convergence**

Regression Loss (Absolute or Squared Distance)

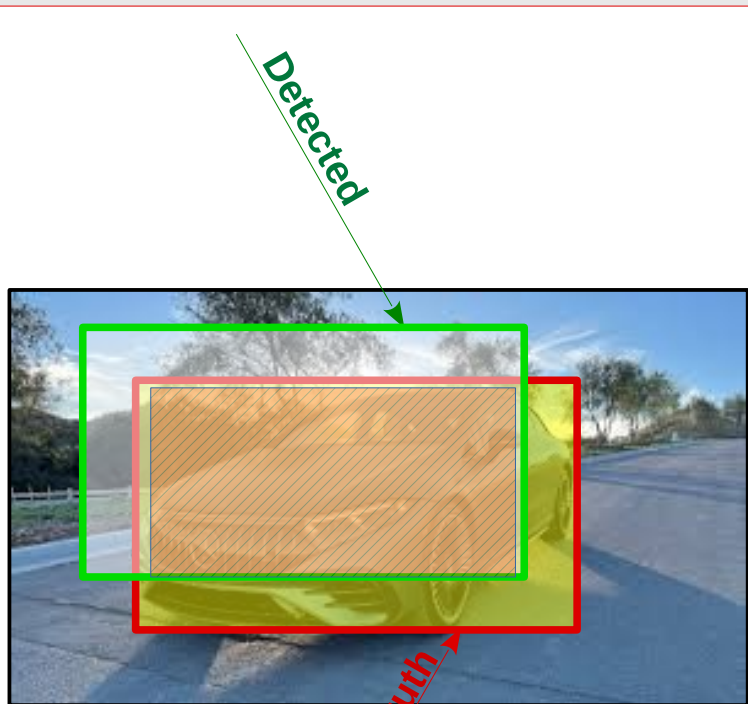


- The prediction for the **larger** object is very close to the ground truth, and
- The prediction for the **smaller** object seems far from the ground truth.
- However, the **distance** of predictions from their ground truth is **the same in both cases**.
- This means that the **loss and the gradient will be the same**.



Intersection over Union

[1] Intersection Over Union (IoU) (Evaluation of Detection Results Accuracy)

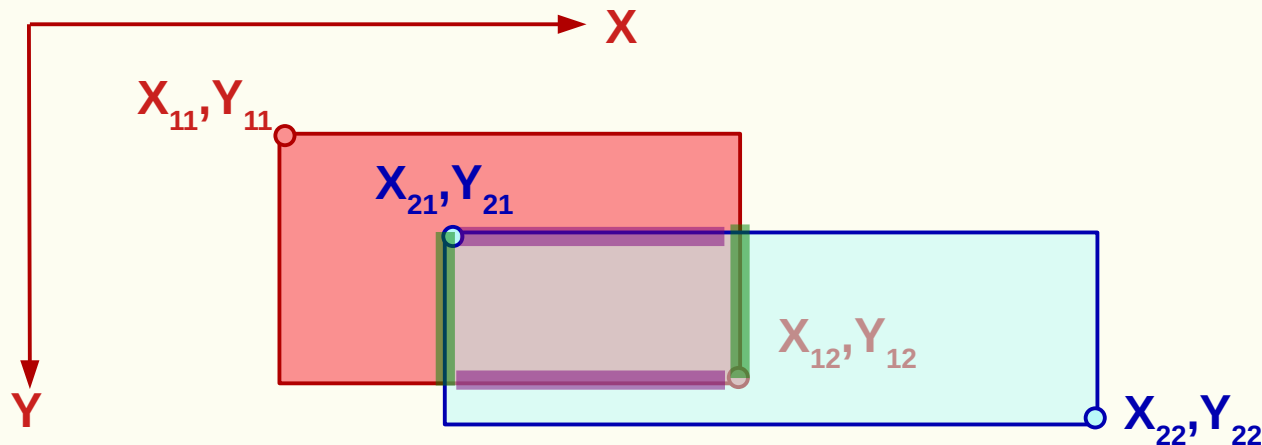


$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$


Acceptance criteria $\text{IoU} > 0.6$ (The higher the better)

IoU is a **measure of overlap** between two bounding boxes

[1] Intersection Over Union (IoU)



$$\text{Intersection Width} = \text{Min} (X_{12}, X_{22}) - \text{Max} (X_{11}, X_{21})$$

$$\text{Intersection Height} = \text{Min} (Y_{12}, Y_{22}) - \text{Max} (Y_{11}, Y_{21})$$

Intersection AREA = Intersection Width * Intersection Height

$$\text{Union AREA} = (X_{12} - X_{11}) * (Y_{12} - Y_{11}) + (X_{22} - X_{21}) * (Y_{22} - Y_{21}) - \text{Intersection AREA}$$

$$\text{IOU} = [\text{Intersection AREA}] / [\text{Union AREA}]$$

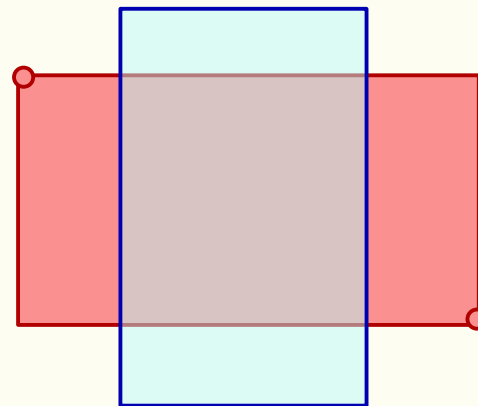
[1] Intersection Over Union (IoU)



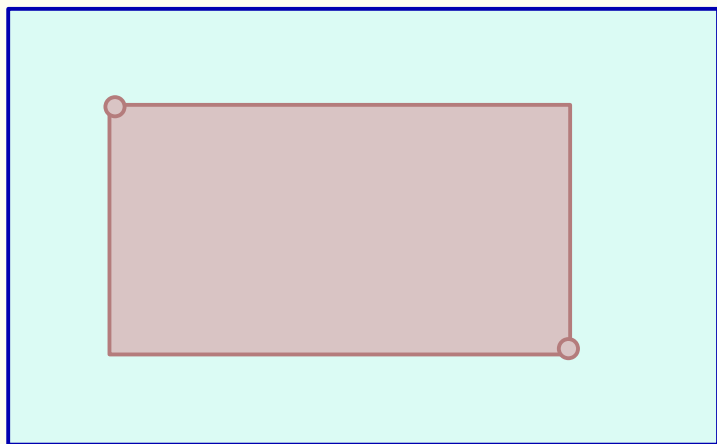
$\text{IOU} < 1$



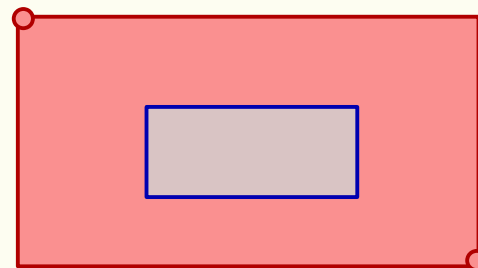
$\text{IOU} = 1$



$\text{IOU} < 1$



$\text{IOU} \ll 1$



$\text{IOU} \ll 1$

Properties of IoU

IoU has several important properties that make it a robust metric for evaluating object detection performance:

- **Non-negativity:** IoU values are always greater than or equal to zero
- **Symmetry:** The value of IoU is the same regardless of the order in which the two shapes are compared
- **Scale Insensitivity:** IoU is invariant to the scale of the shapes being compared, making it a versatile metric

[1] IOU-Based Loss function

The Higher the IOU, the Better the performance

It is required to **Maximize** the **IOU** of the model output

It is required to **Minimize** (**1-IOU**) of the model output

Minimize the IOU Loss using Gradient Descent

$$Loss_{IOU} = 1 - IOU$$

[1] IOU-Based Loss function

Note:

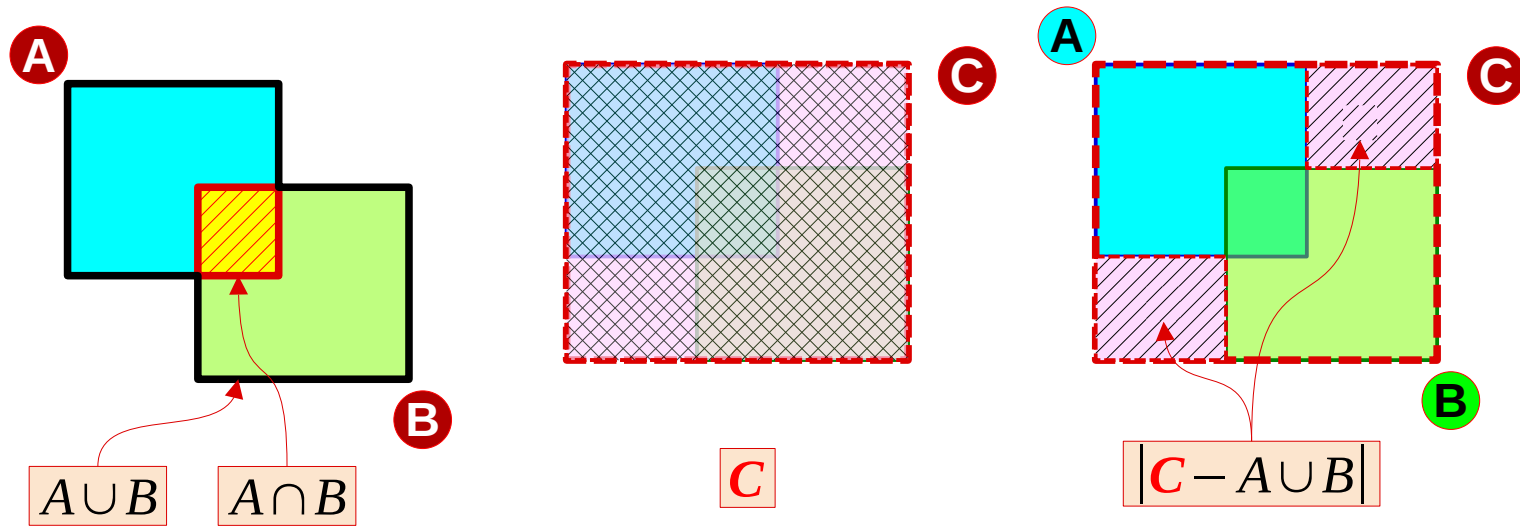
- If there are no overlaps between ground truth and prediction, the IoU will be zero
- The loss = $1 - \text{IoU} = 1$. which is constant.
- With no change in loss, the gradient will be zero.
- Zero Gradient leads to NO Update in Model weights
- This is typical the case with initial prediction during the training

$$\text{Loss}_{\text{IoU}} = 1 - \text{IoU}$$

How to enhance the IoU-Based Loss Function ?

[2] GIoU (Generalized IoU) Loss Function

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C: the smallest **convex** object that encloses **A** and **B**
for example: **smallest rectangle** that enclose both **A** and **B**

$$IoU = \frac{|A \cap B|}{|A \cup B|}$$

$$GIoU = IoU - \frac{|C - A \cup B|}{|C|}$$

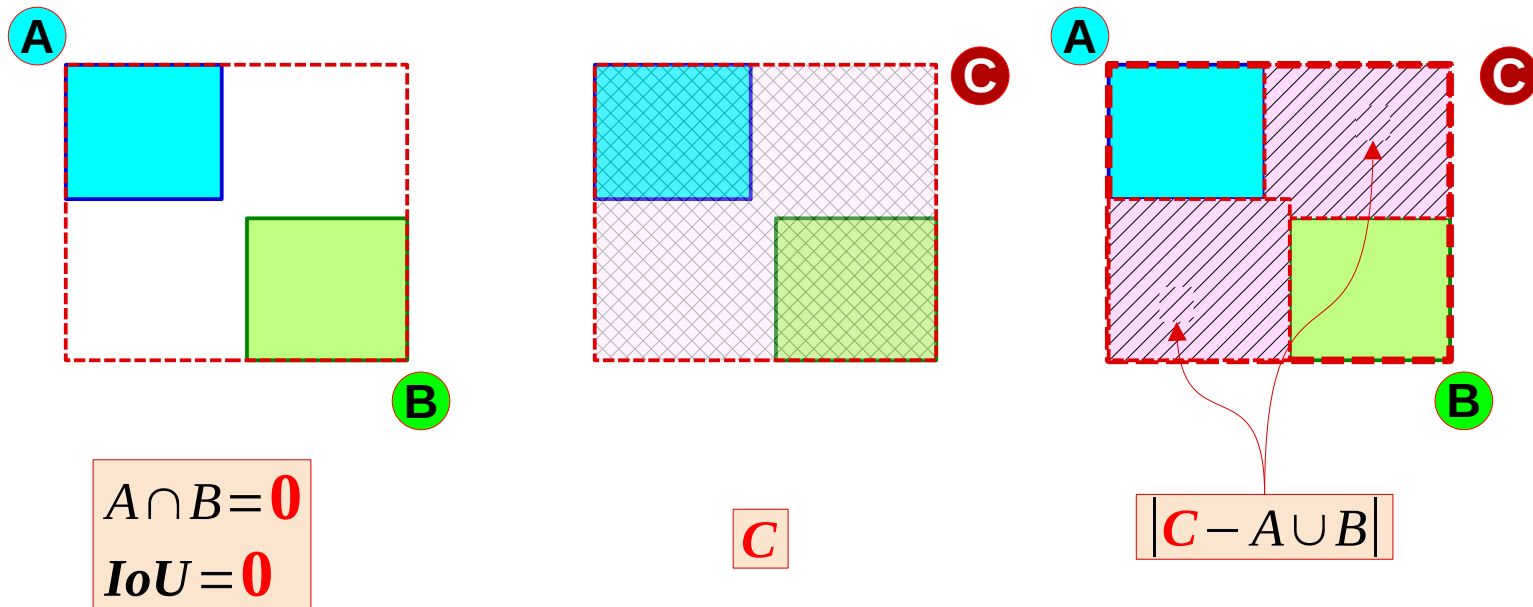
$$Loss_{IoU} = 1 - GIoU$$

$$Loss_{GIoU} = 1 - IoU + \frac{|C - A \cup B|}{|C|}$$

[2] GIoU (Generalized IoU) Loss Function

Non-Overlap Case)

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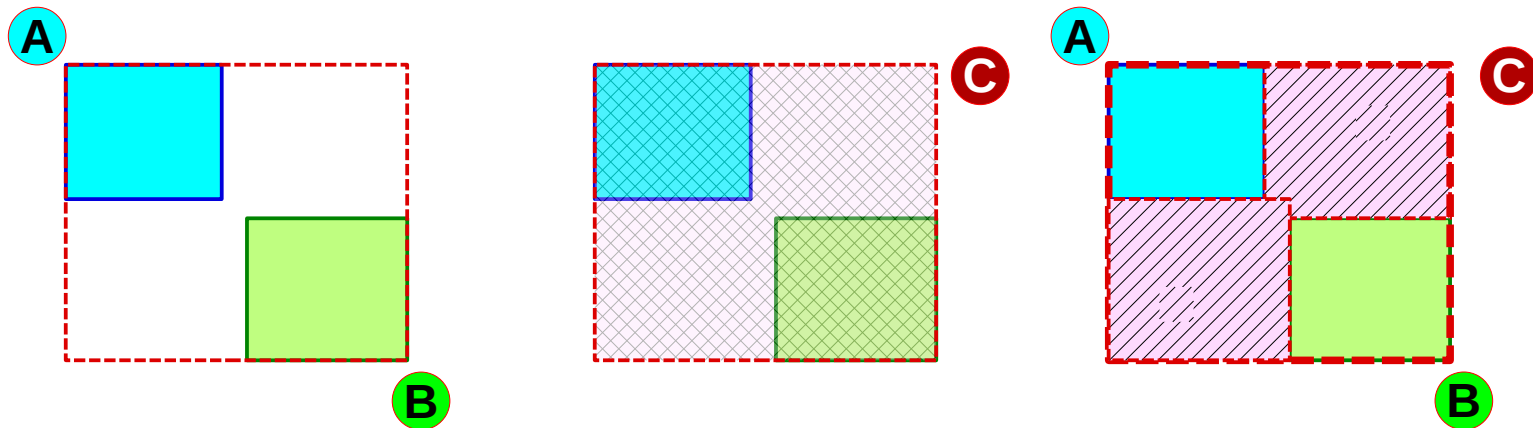
$$IoU = \frac{|A \cap B|}{|A \cup B|}$$

$$GIoU = IoU - \frac{|C - A \cup B|}{|C|}$$

$$Loss_{IoU} = 1 - GIoU$$

$$Loss_{GIoU} = 1 - IoU + \frac{|C - A \cup B|}{|C|}$$

[2] GIoU (Generalized IoU) Loss Function (Non-Overlap Case)



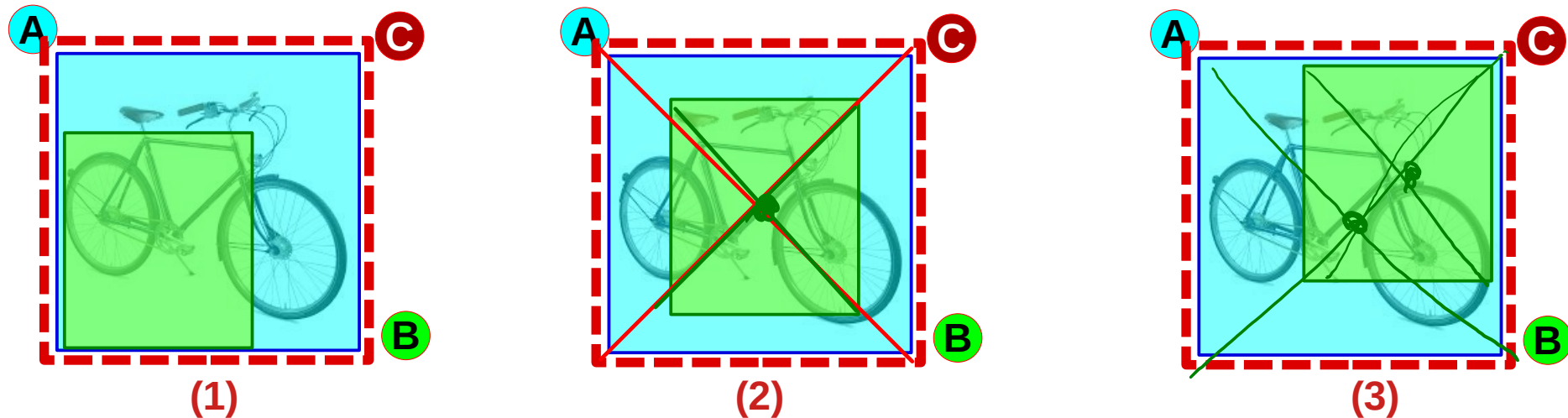
$$A \cap B = 0$$
$$IoU = 0$$

$$Loss_{GIoU} = 1 - IoU + \frac{|C - A \cup B|}{|C|}$$

GIoU depends on how far the prediction and ground truth boxes are.
Even if **IoU** is zero, *it has a gradient* to push the bounding box toward the ground truth.

[2] Problem with GloU (Generalized IoU) Loss Function

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Same A, B

Same IoU

Same C

Same $G IoU$

$$IoU = \frac{|A \cap B|}{|A \cup B|}$$

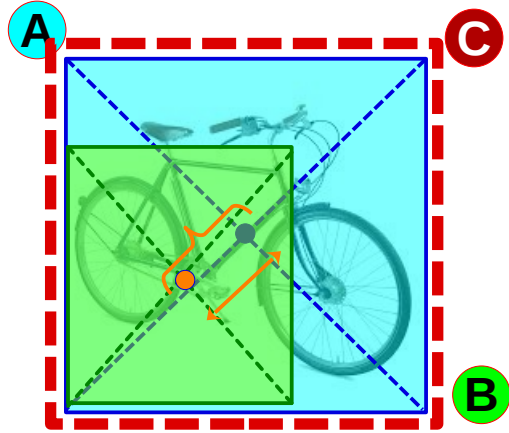
$$Loss_{G IoU} = 1 - IoU + \frac{|C - A \cup B|}{|C|}$$

Prediction number (2) is the best since its center coincide with the center of the ground truth

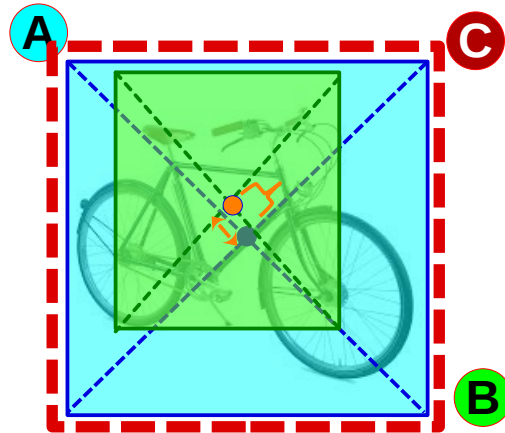
GIoU does NOT take into consideration the distance between centers of Ground Truth and Prediction

[3] DIoU (Distance IoU) Loss Function

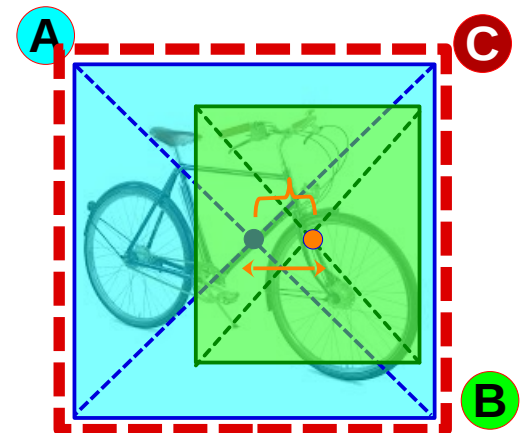
18



(1)



(2)



(3)

Where:

d^2 : Squared distance between the **two centers**

C^2 : Squared length of **diagonal** of **C**

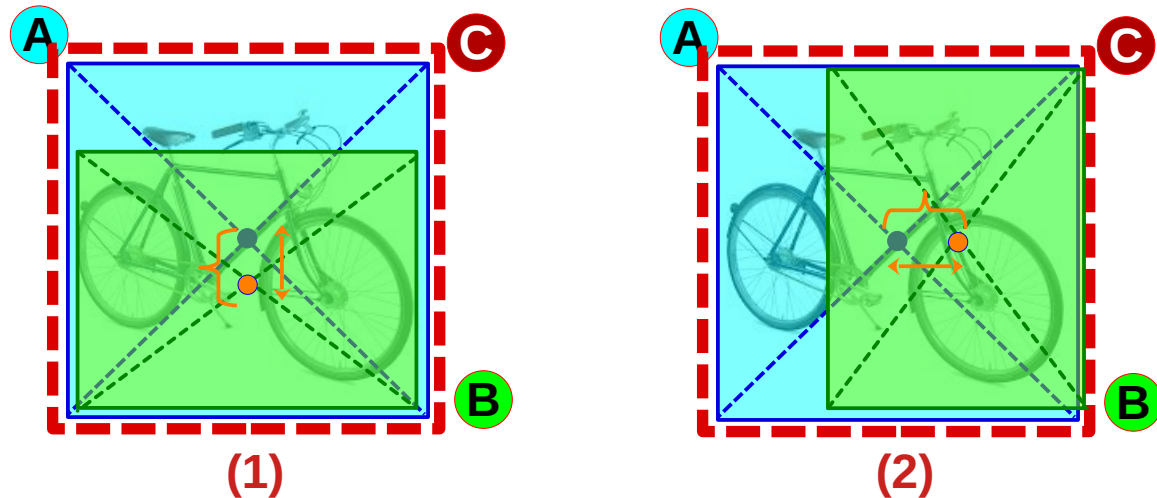
$$IoU = \left| \frac{A \cap B}{A \cup B} \right|$$

$$DIOU = IoU - \frac{d^2}{C^2}$$

$$Loss_{DIOU} = 1 - IoU + \frac{d^2}{C^2}$$

[3] Problem with DIoU (Distance IoU) Loss Function

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Problem:

d^2 : **SAME** Squared distance between the **two centers**

C^2 : **SAME** Squared length of **diagonal** of **C**
SAME IoU

Box(1) is better than **Box**(2)

$$IoU = \left| \frac{A \cap B}{A \cup B} \right|$$

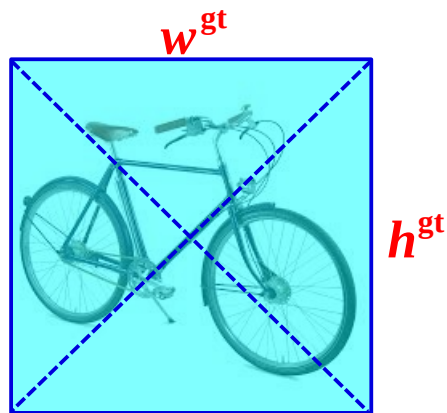
$$DIOU = IoU - \frac{d^2}{C^2}$$

$$Loss_{DIOU} = 1 - IoU - \frac{d^2}{C^2}$$

Aspect Ratios of boxes are **NOT** taken into consideration

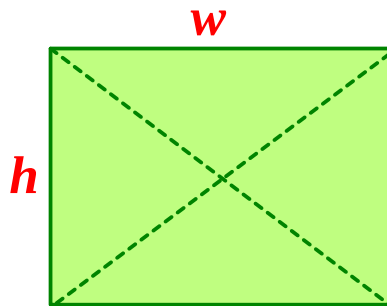
[4] CIoU (Complete IoU) Loss Function (Measure of Orientation)

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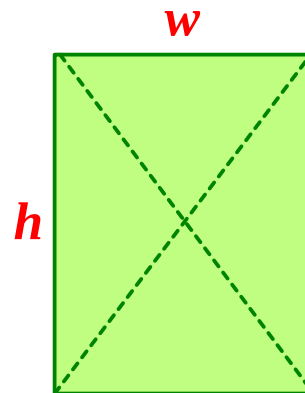


Ground Truth

$$angle = \arctan \frac{w^{gt}}{h^{gt}}$$



$$angle = \arctan \frac{w}{h}$$



$$angle = \arctan \frac{w}{h}$$

angle range
(0 to $\pi/2$)

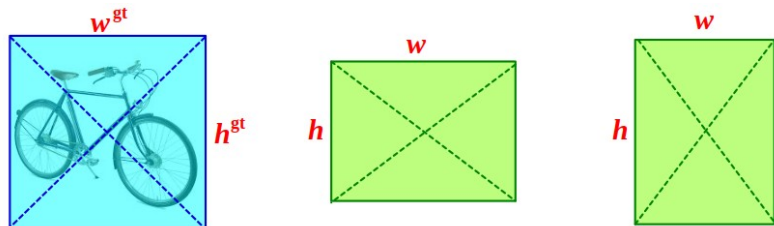
$$Angle Distance = \left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h} \right)^2$$

Angle Distance is a measure of **Orientation**

$$Normalized Angle Distance = \left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h} \right)^2 / (\pi/2)^2 = \frac{4}{\pi^2} \left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h} \right)^2$$

[4] CIOU (Complete IoU) Loss Function (Modified DIOU)

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$$IoU = \left| \frac{A \cap B}{A \cup B} \right|$$

$$DIOU = IoU - \frac{d^2}{C^2}$$

$$\text{Orientation } \nu = \frac{4}{\pi^2} \left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h} \right)^2$$

$$CIOU = IoU - \frac{d^2}{C^2} - \alpha \nu$$

Where α is **orientation** weight factor based on IOU $\alpha = \frac{\nu}{(1 - IoU) + \nu}$

$$Loss_{CIOU} = 1 - IoU + \frac{d^2}{C^2} + \alpha \nu$$

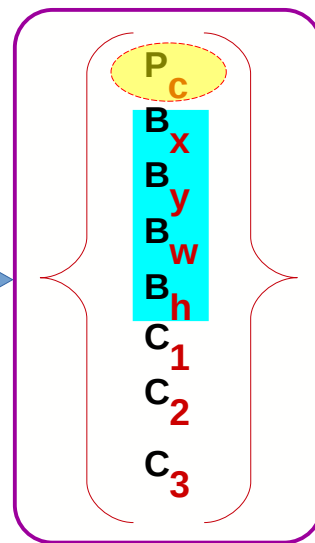
- **EIoU** (Efficient IoU)
- **Focal-EIoU**
- **SIoU** (SCYLLA-IoU)
- **WIoU** (Wise-IoU)

Reference: <https://medium.com/@cshyo1004/iou-and-variants-overview-a328acf177cd>

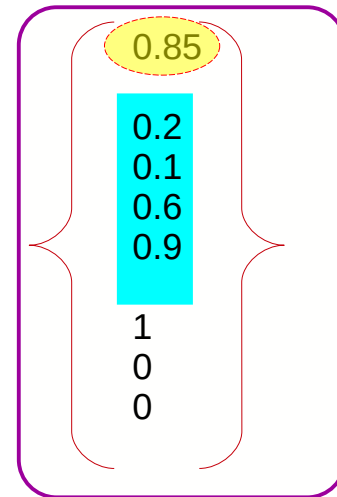
Selection of the **Best Detected** Object bounding box

Review: Localization and Classification

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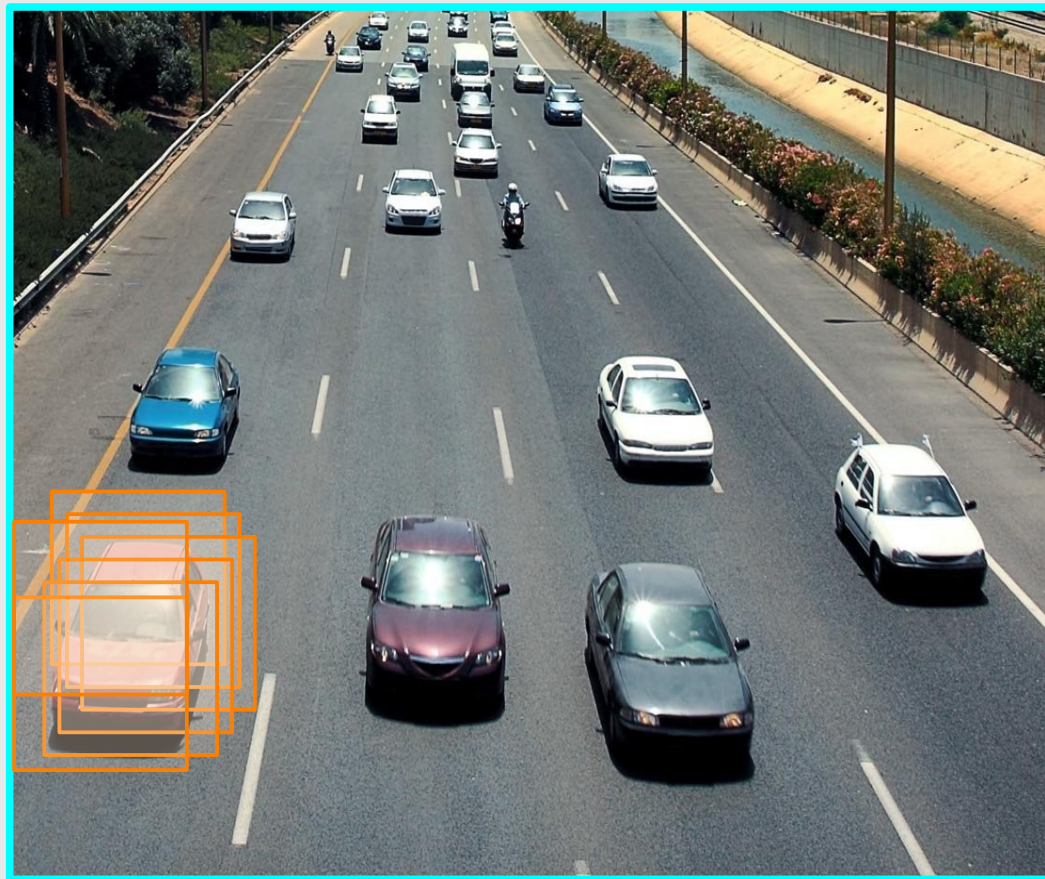


Output Y



Label for
Car

Problem of OVERLAPPED Detections



**Same object
can be detected with
Multiple windows !!!**

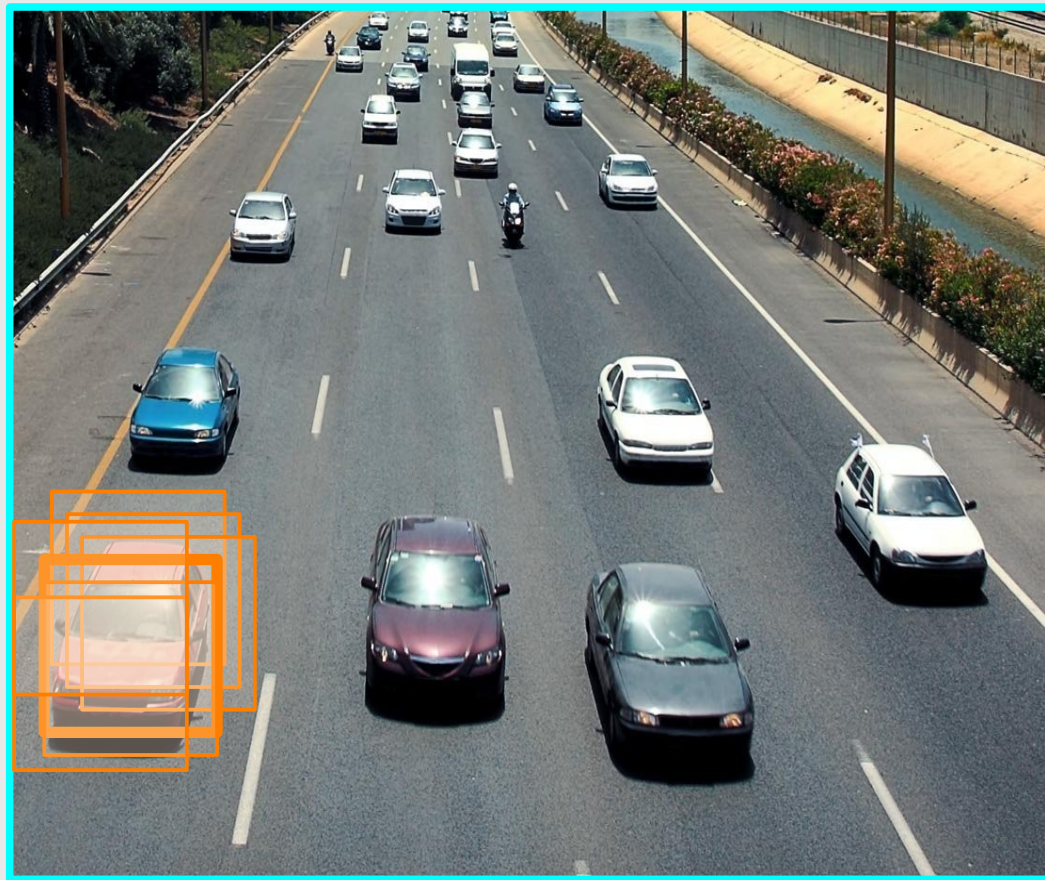
Which Bounding box is the best ??
(Encloses most of the object)

Remember:

Each Box has its own Probability P_c

The higher the **probability** the more **Confidence** level

Problem of OVERLAPPED Detections



**Same object
can be detected with
Multiple windows !!!**

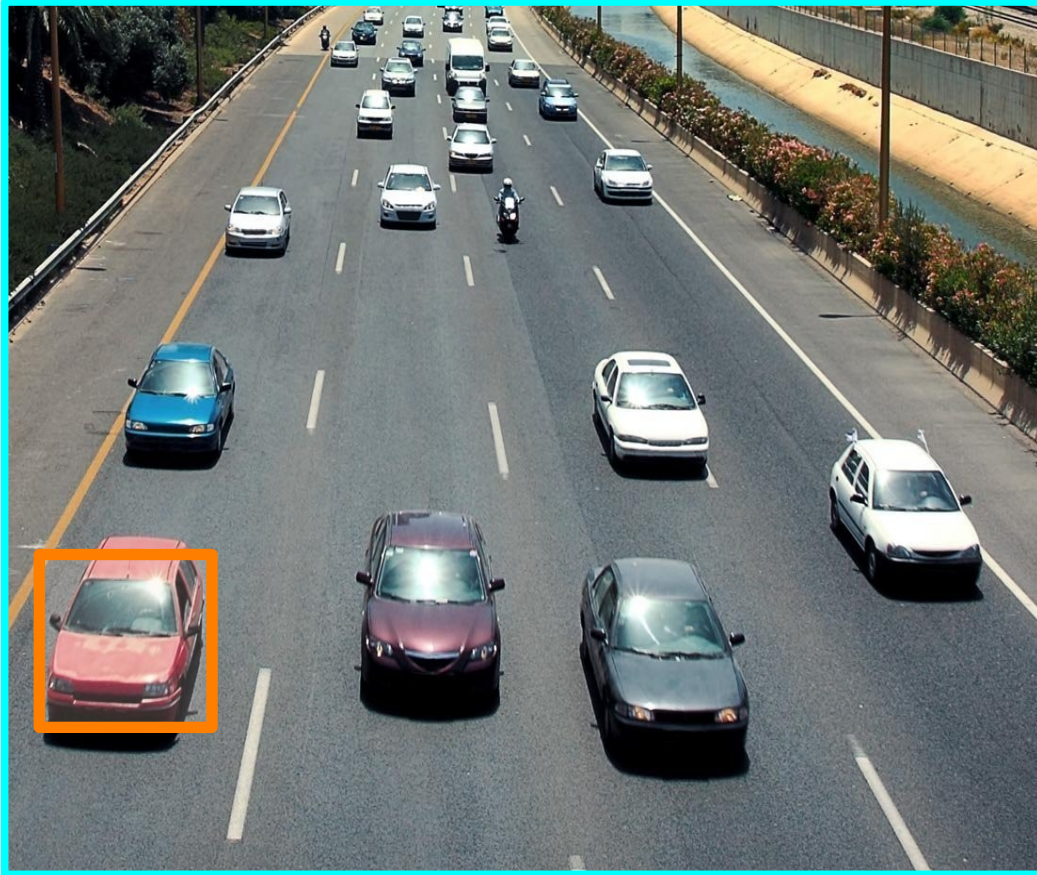
Solution:
*Select the Box with highest Confidence
Discard Other Boxes*

Remember:

Each Box has its own Probability P_c
The higher the **probability** the more **Confidence** level

Problem of OVERLAPPED Detections

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**Same object
can be detected with
Multiple windows !!!**

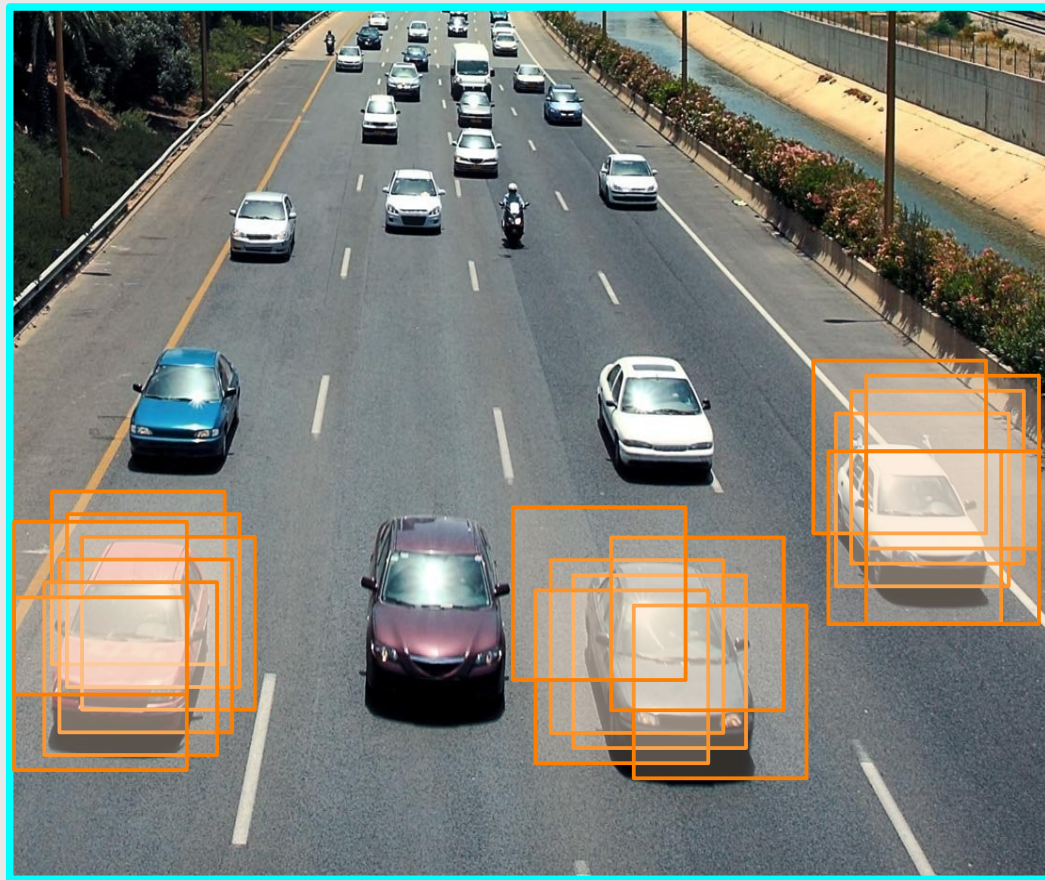
Solution:
Select the Box with highest Confidence
Discard Other Boxes



Remember:

Each Box has its own Probability P_c
The higher the **probability** the more **Confidence** level

Problem of OVERLAPPED Detections

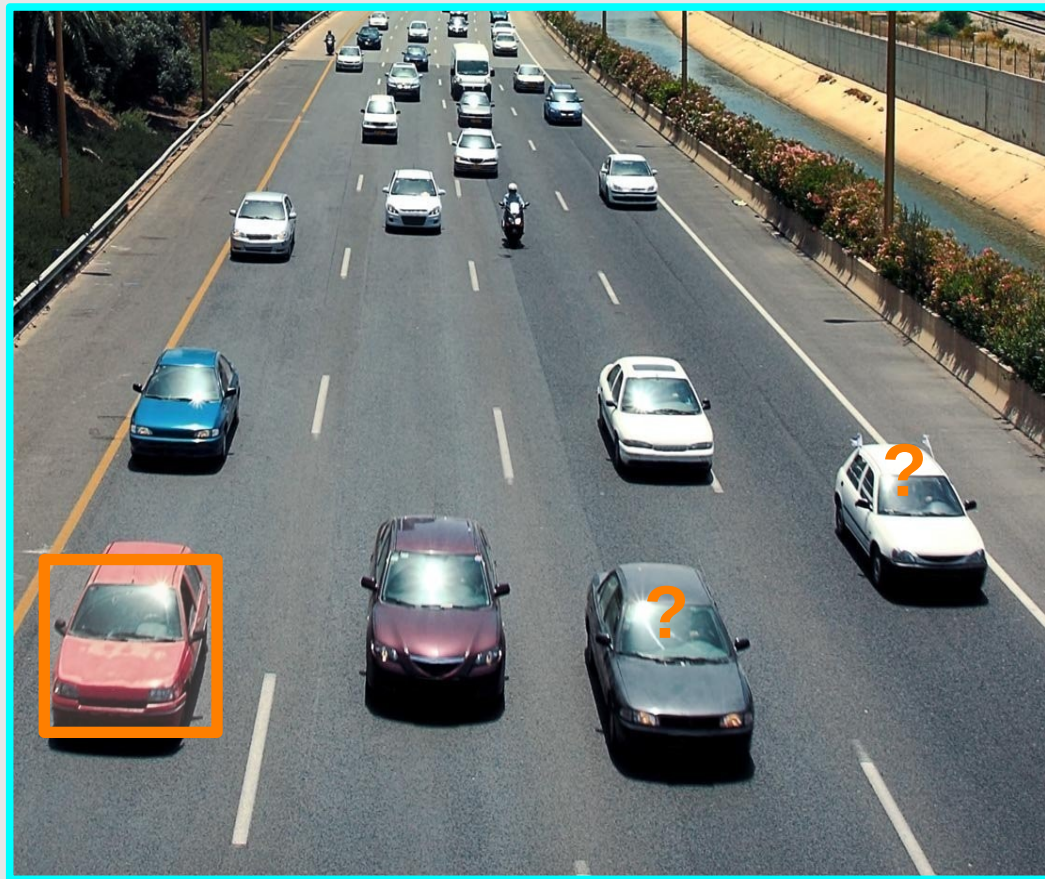


**Same object
can be detected with
Multiple windows !!!**

Solution:
Select the Box with highest Confidence
Discard Other Boxes

Multiple Objects are detected in the same image

Problem of OVERLAPPED Detections



Same object
can be detected with
Multiple windows !!!

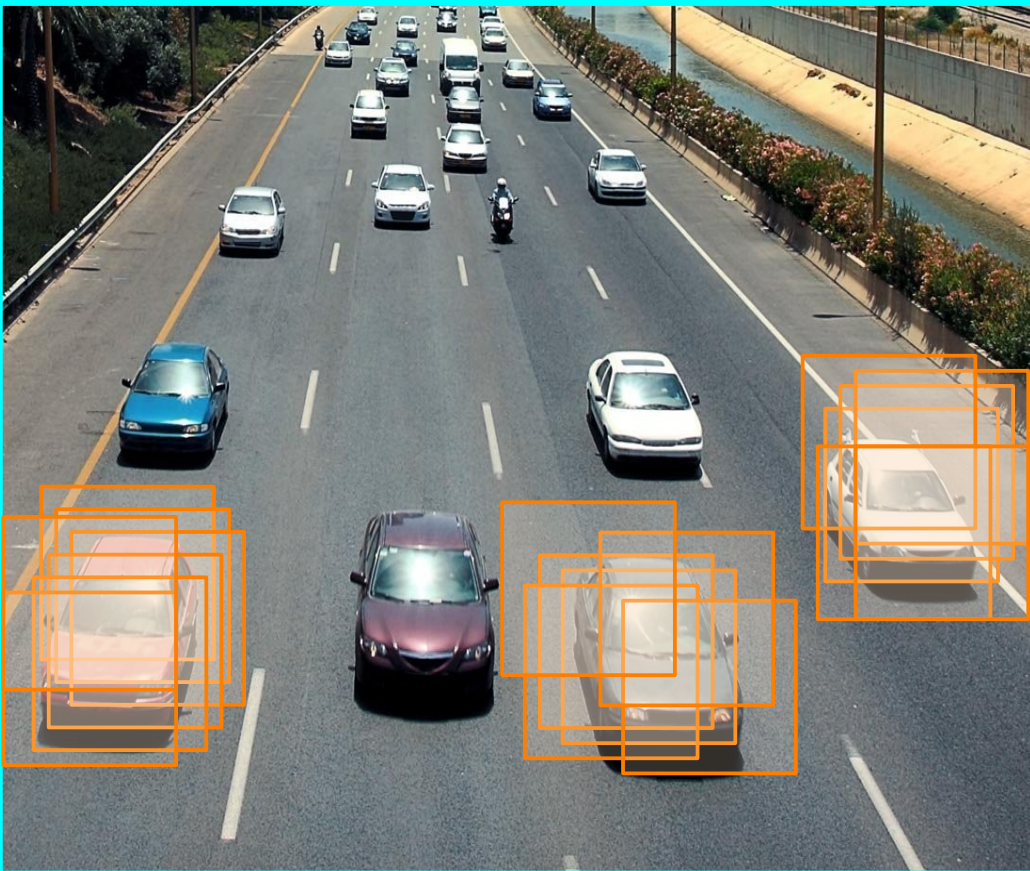
Solution:
Select the Box with highest Confidence
Discard Other Boxes



Multiple Objects are detected in the same image

Non-Maximum Suppression Algorithm

Non-Max Suppression

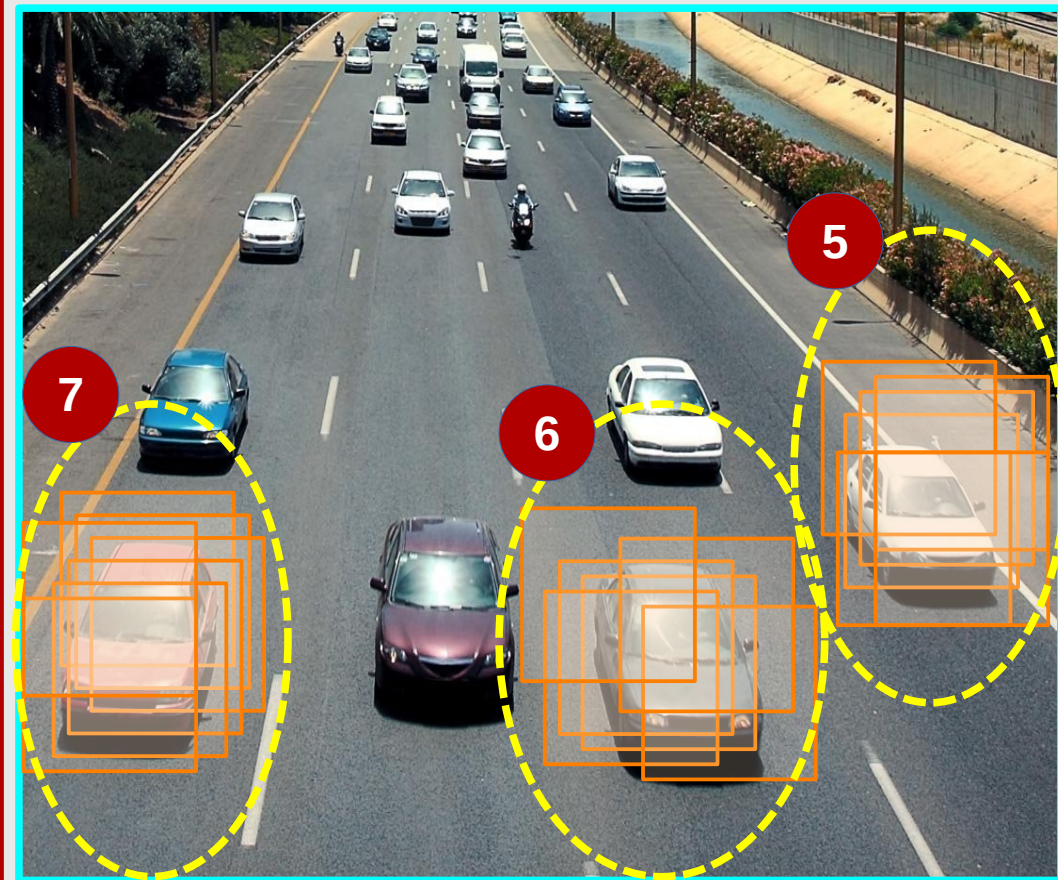


Note: $P_c < 0.6$ and $IoU > 0.5$ are user defined

Non-Max Suppression Algorithm

- 1- Discard All Bounding Boxes with $P_c < 0.6$
[Low probability to be an Object]
- 2- Select One Box with **Highest P_c**
- 3- Calculate **IOU** between **selected box** and **ALL other Boxes**
- 4- **Discard** Boxes with **IOU > 0.5**
(Highly Overlapped with selected box)
- 5- **Repeat** 2,3,4 excluding the previously selected predicted Box
- 6- **Stop** when there is **no Box to discard**

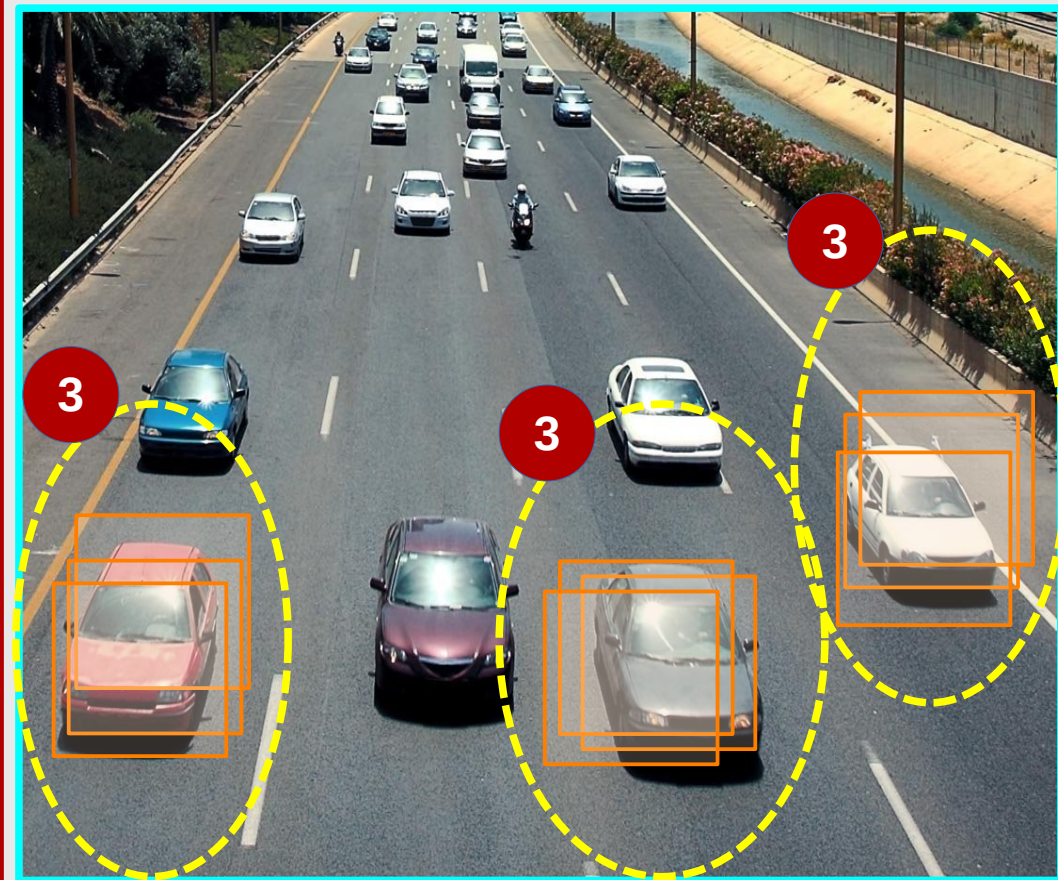
Non-Max Suppression – Step By Step



Original Output from the Detection Model

Note: $P_c < 0.6$ and $IoU > 0.5$ are user defined

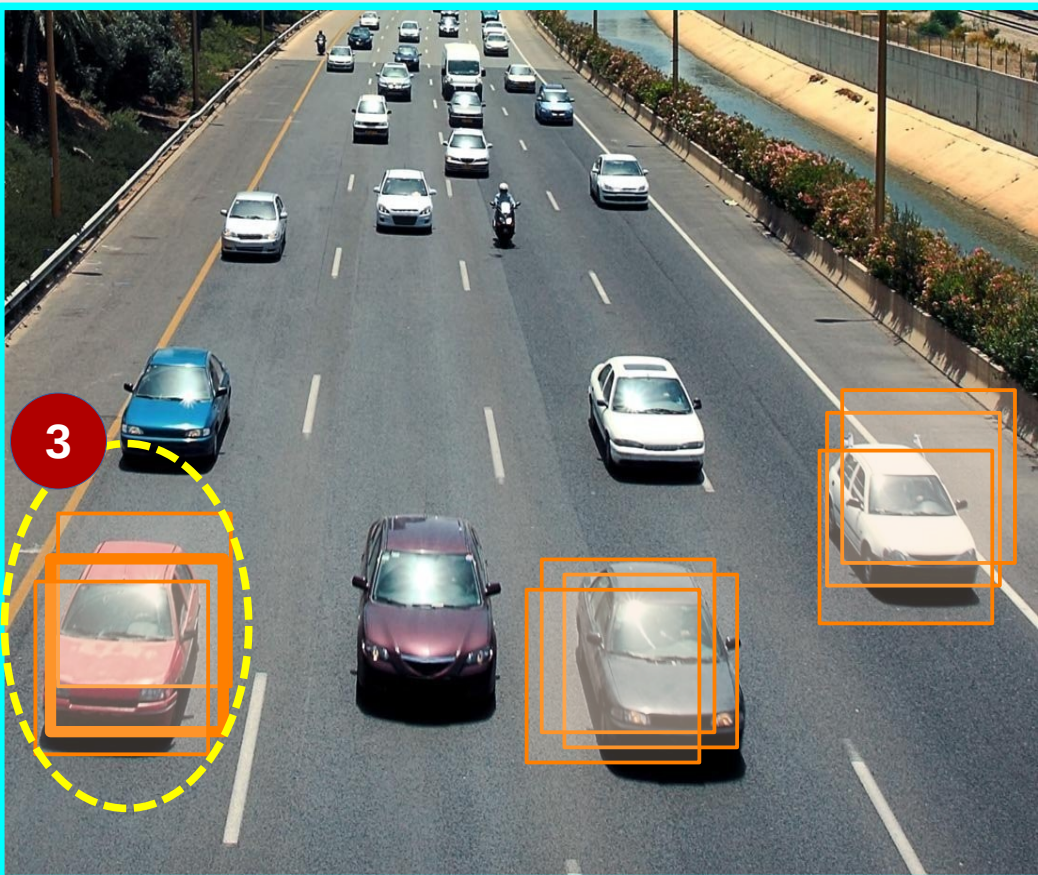
Non-Max Suppression



1- Discard All Bounding Boxes with $P_c < 0.6$
[Low probability to be an Object]

Note: $P_c < 0.6$ and $IoU > 0.5$ are user defined

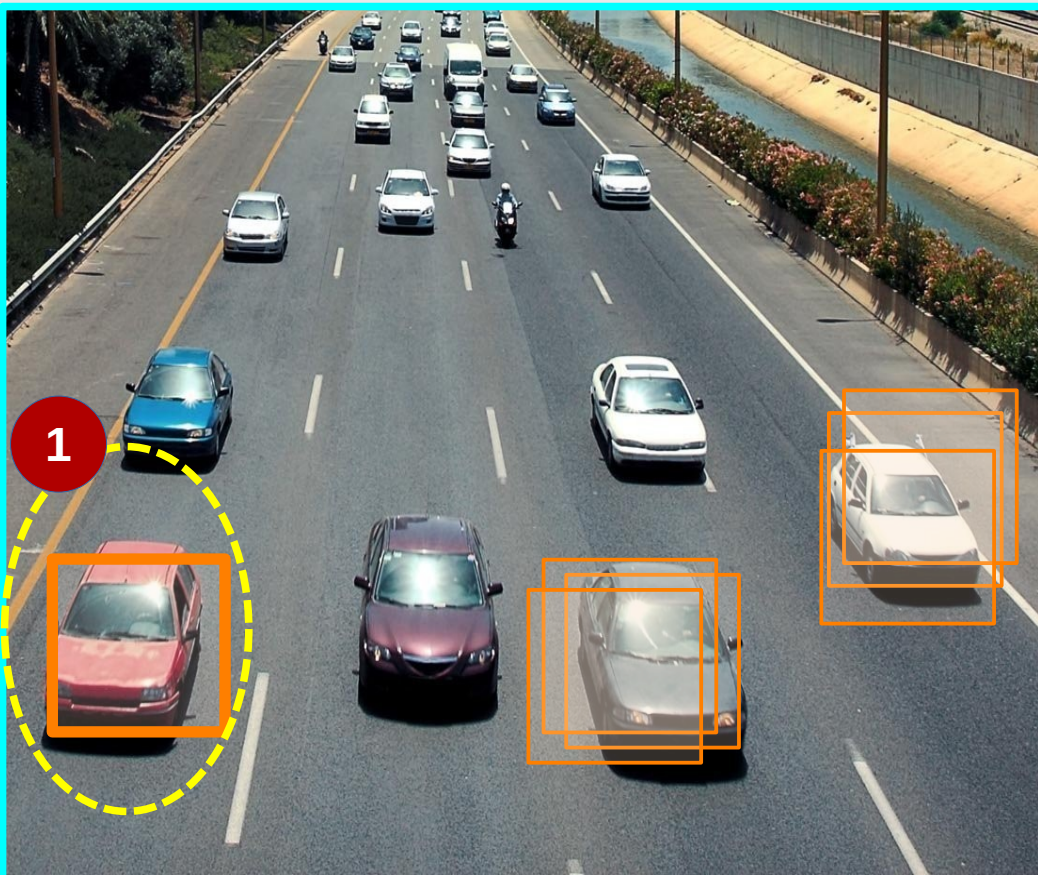
Non-Max Suppression



2- Select One Box with **Highest P_c**
(Over all bounding Boxes)

Note: $P_c < 0.6$ and $IoU > 0.5$ are user defined

Non-Max Suppression

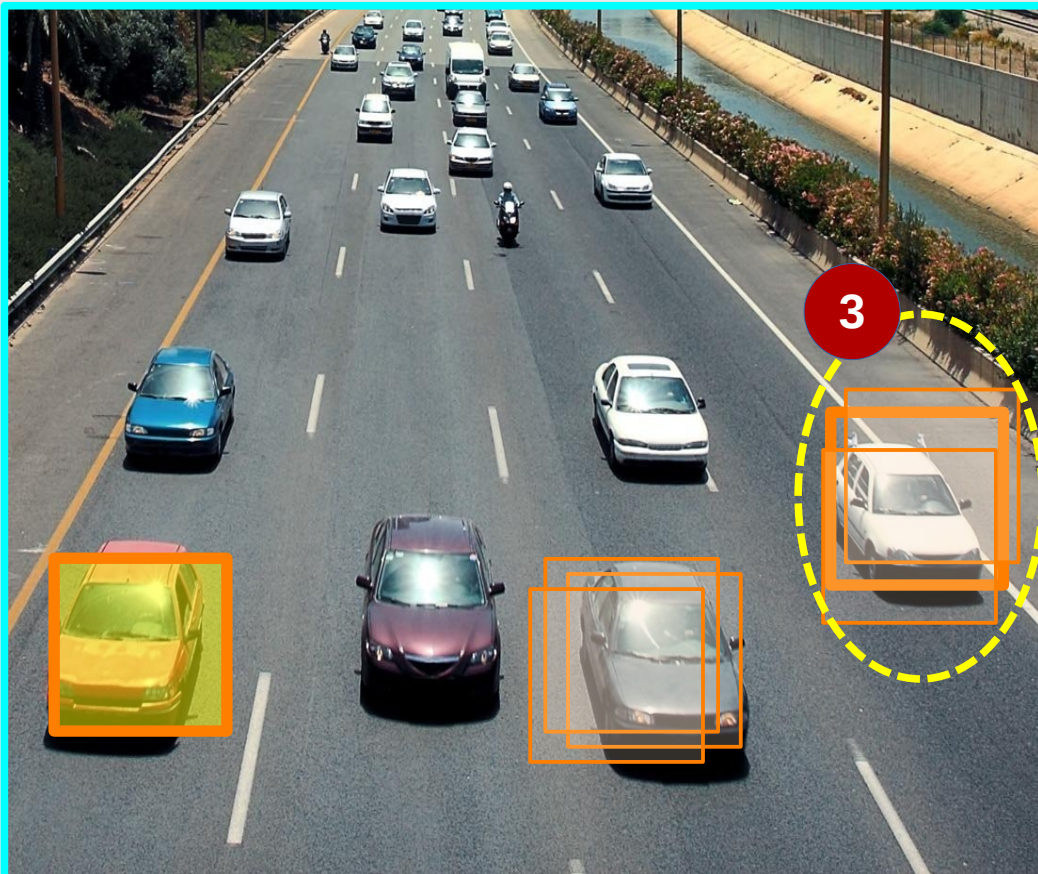


3- Calculate **IOU** between selected box and ALL other Boxes

4- **Discard** Boxes with **IOU > 0.5**
(Highly Overlapped with selected box)

Note: $P_c < 0.6$ and $IoU > 0.5$ are user defined

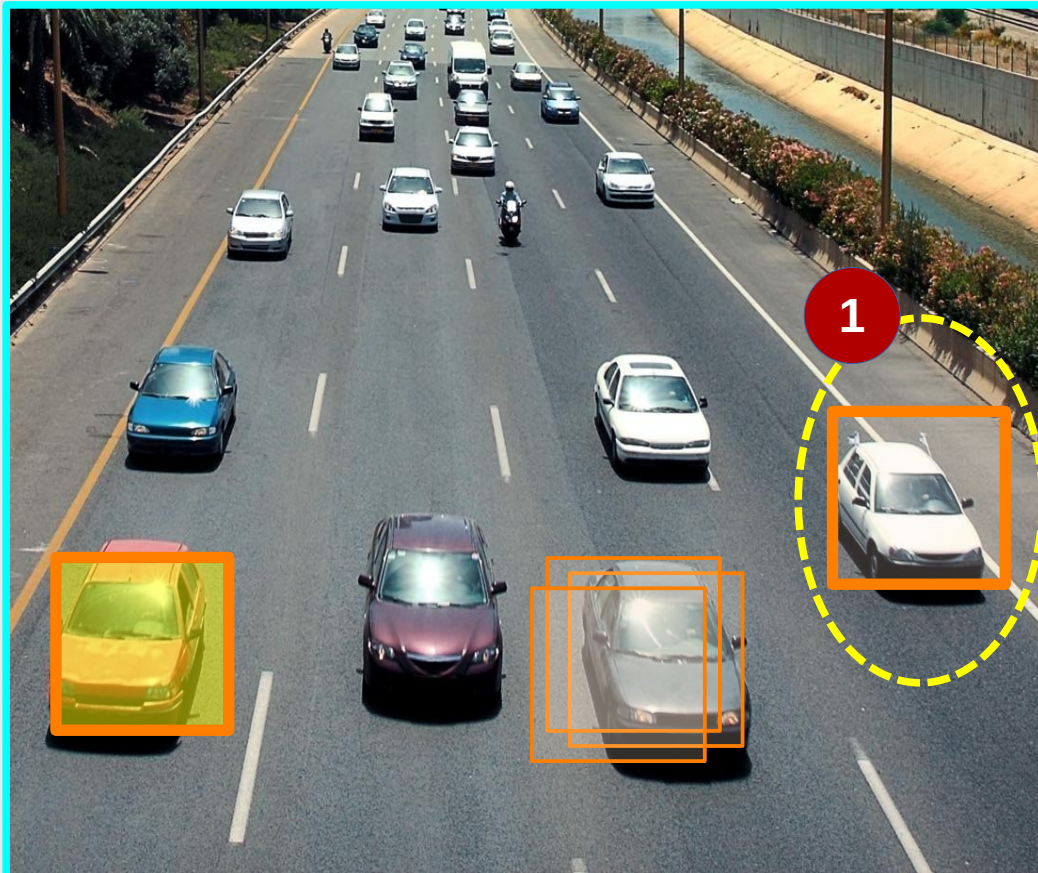
Non-Max Suppression



2- Select One Box with **Highest P_c**
(Over all bounding Boxes)

Remember: Selected Boxes are **Excluded**

Non-Max Suppression

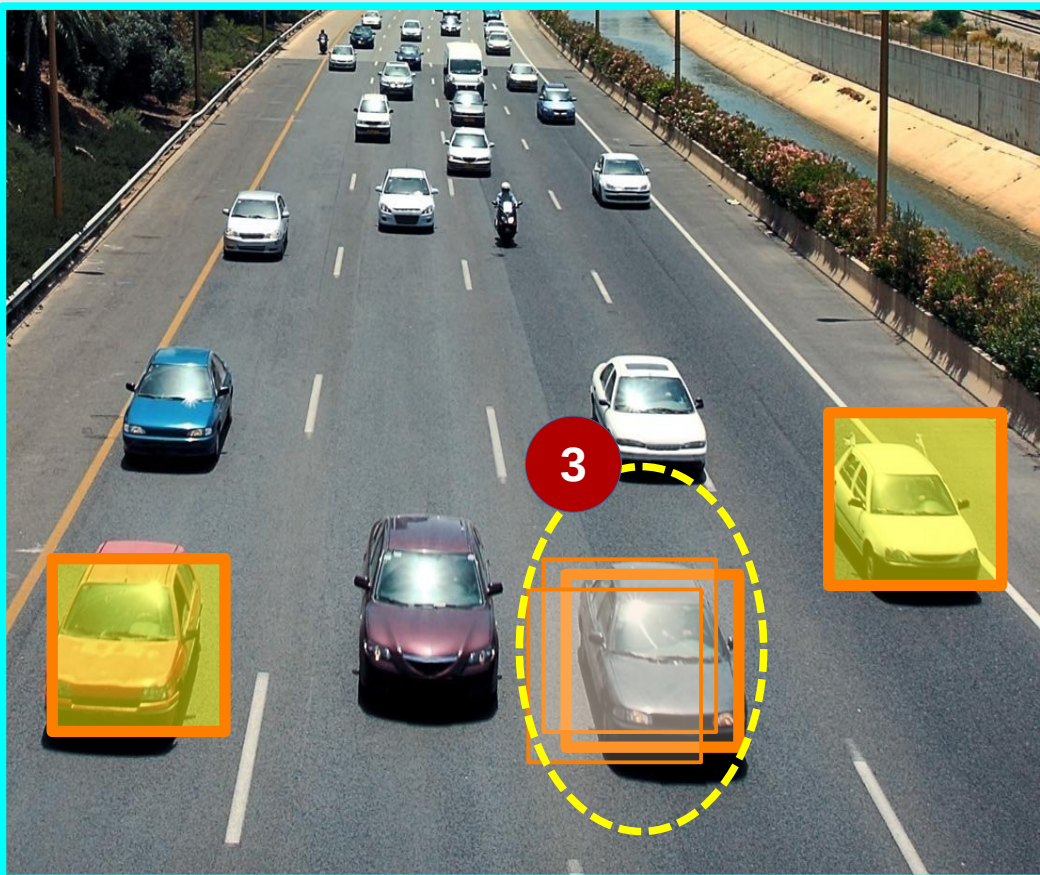


3- Calculate **IOU** between selected box and ALL other Boxes

4- **Discard** Boxes with **IOU > 0.5**
(Highly Overlapped with selected box)

Remember: Selected Boxes are **Excluded**

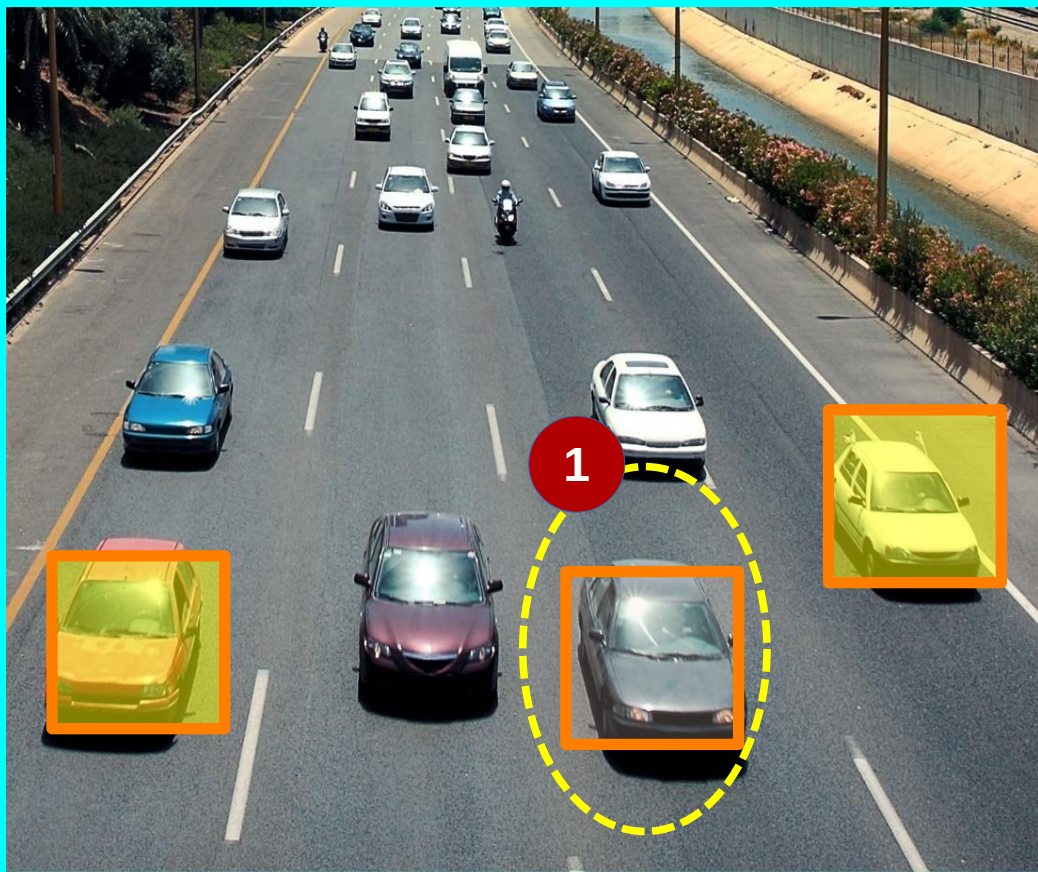
Non-Max Suppression



2- Select One Box with **Highest P_c**
(Over all bounding Boxes)

Remember: Selected Boxes are **Excluded**

Non-Max Suppression

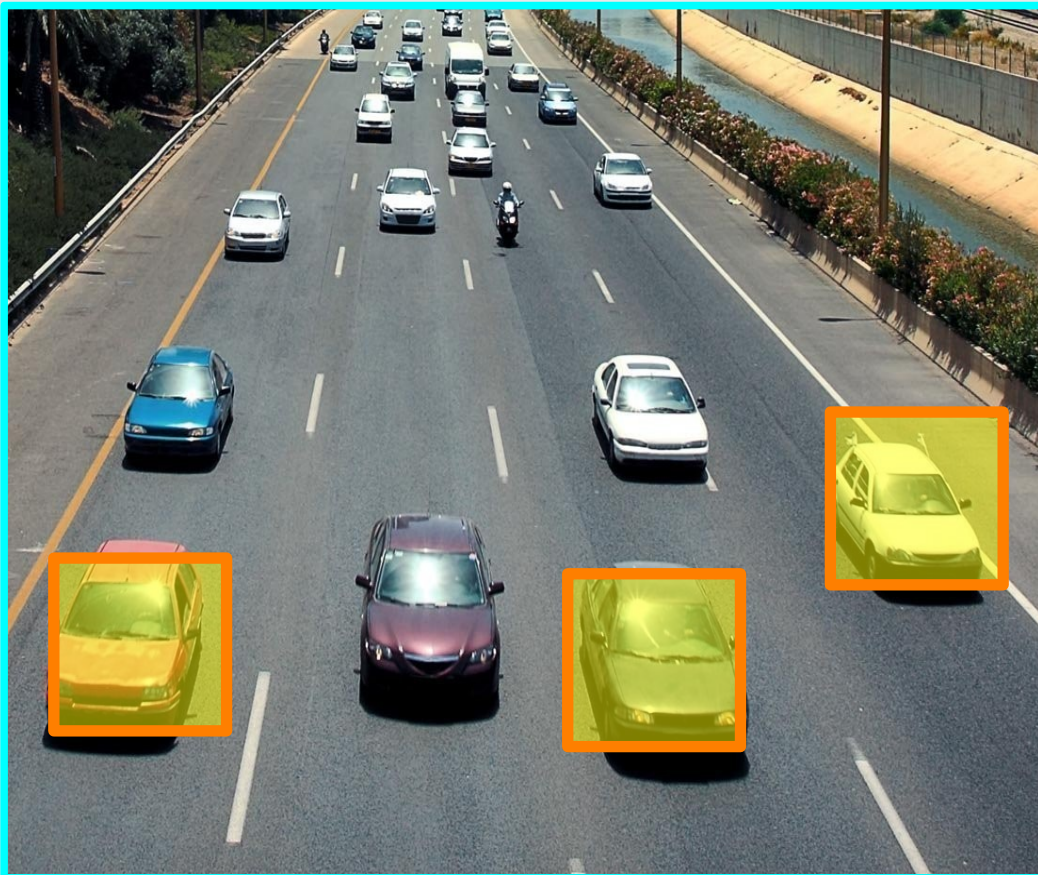


3- Calculate **IOU** between selected box and ALL other Boxes

4- **Discard** Boxes with **IOU > 0.5**
(Highly Overlapped with selected box)

Remember: Selected Boxes are **Excluded**

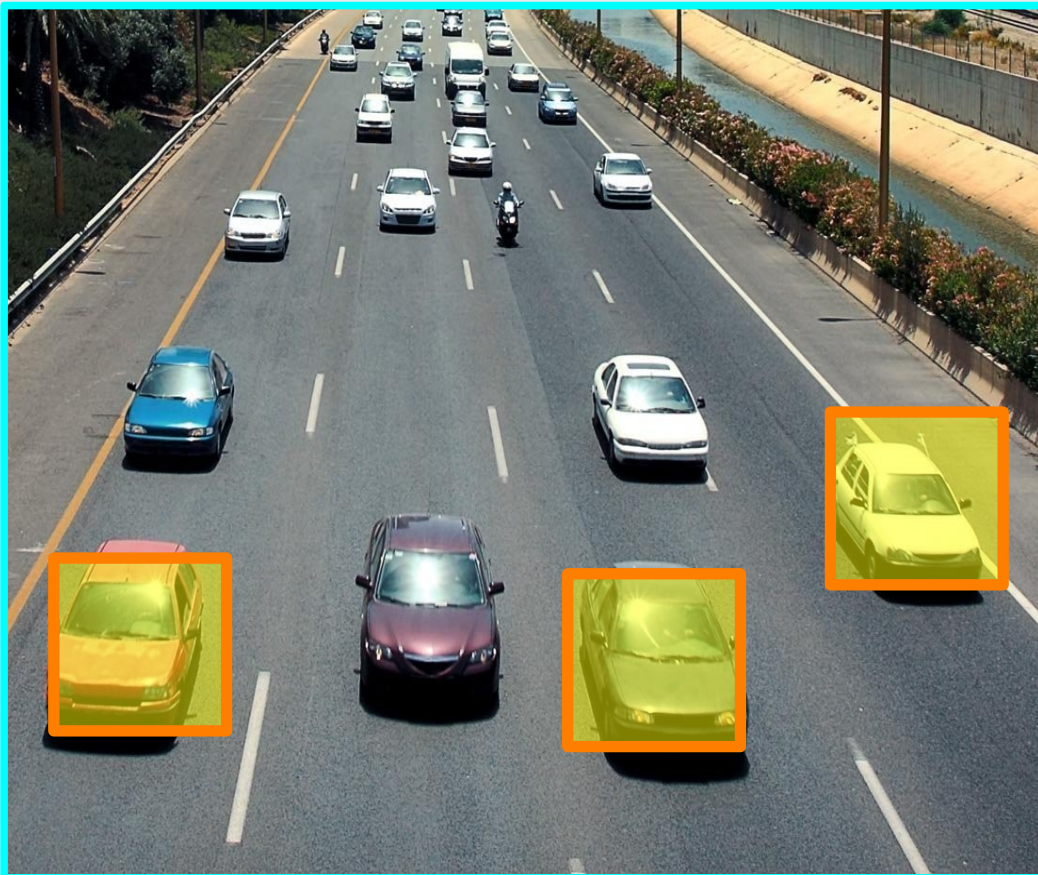
Non-Max Suppression



6- **Stop** when there is **no Box to discard**

Remember: Selected Boxes are **Excluded**

Non-Max Suppression



Summary:

NMS is a Greedy Algorithm

At Each Iteration, Pick the highest Scoring Box

Remove ALL boxes that are overlap with the selected Box

Remember: Selected Boxes are Excluded